Title of the Thesis

Bachelor/Master Thesis Proposal

Firstname Lastname

February 1, 2018

1 Motivation

1.1 Research field

General description of the research field (for example, VANETs, IMD, Simulation), discussing characteristics and applications, including some introductory literature as appropriate. **(this page)**

...

1.2 Thesis area

Introduce the specific topic of the work here (e.g., Communication protocols, security, privacy, misbehavior detection) and describe the requirements and goals of the work, such as the required efficiency. **(Half a Page)**

...

2 State of the Art

Initial literature related to the specific topic. This will be a small selection of papers relevant to the work, agreed on with the supervisor. The purpose of this section is to demonstrate problems or limitations of existing solutions. **(1 page)**

...

3 Problem Statement

3.1 Thesis focus

Based on the Motivation (1) and existing work (2) this section should describe the problems to be solved in this work. This should include a description of the additional ideas that will be designed (e.g., a concept, implementation, ...) and the evaluation methodology (e.g., formal analysis, simulation, user study). **(0,75 pages)**

...

3.2 Research questions

Research questions describe one or several main questions of the thesis, as well as several sub-questions agreed upon with the supervisor. Research questions can be split in several types and are typically W-questions (why, what, how, ...).[a]

Explanation: Why is something the way it is?

Description: What do different approaches look like?

Prediction: How will something develop?

Assessment: How can the described/explained be assessed?

These questions should be written such that they can be answered within the thesis, and they should fully cover the research problem. **(0,25 Pages)**

[a] based on `http://www.studieren.at/uni-abc/forschungsfrage-welche-fragetypen-gibt-es` (German).

Main question ...

1. Partial question
2. Partial question
3. ...

4 Approach

Describe the initial ideas for the thesis and the first steps to answer the research questions. Ideas can be pointers towards promising ideas in the literature (not necessarily read completely) or some sketch of how ideas from the literature may be developed. It does not need to be a concrete idea, but should indicate what will be done in the first weeks after the proposal is done. **(0,5 Pages)**

...

5 Planning

5.1 Own Background

Relevant background on methodologies and tools (e.g., Simulator, appropriate models, algorithms) from lectures, seminars and jobs. Also write down the necessary skills that need to be learned during the thesis. **(0,25 Pages)**

...

5.2 Required Resources

If applicable, a list of required hardware or other resources. **(arbitrary length)**

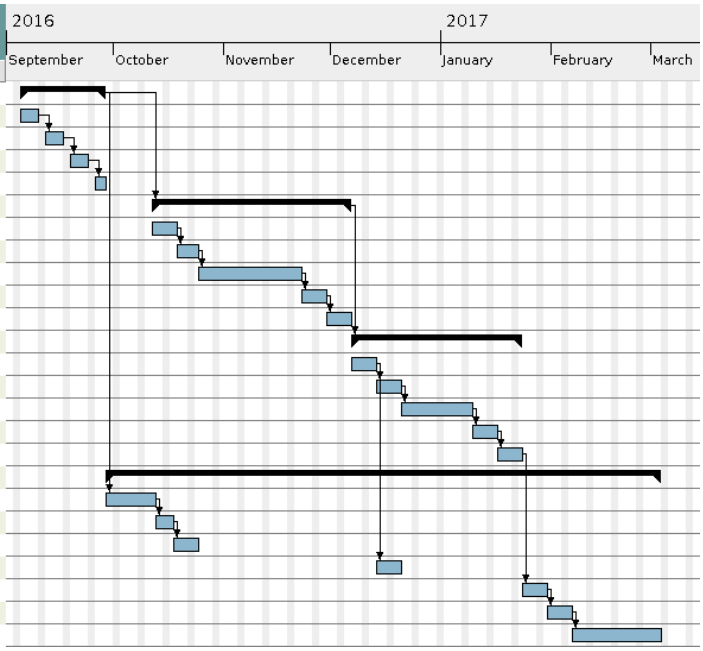
Resource description...

Resource description...

5.3 Work packages

General description of the work. For the first months of the thesis, the plan should be on a weekly bases. Later workpackages can be more coarse. The plan may be adapted during the course of the thesis, if required. However, the initial plan must include necessary steps for the completion of the thesis. The time plan should be consistent with the research questions (3.2) and the necessary background (5.1) and potential wait time for resources (5.2). The plan may be included as a gantt chart (see following example), or in any other desired form. **(0,5 Pages)**

Name	Begin ...	End date
Researching suitable tools	Sep, 5	Sep, 28
Testing tool A	Sep, 5	Sep, 9
Testing tool B	Sep, 12	Sep, 16
Testing tool C	Sep, 19	Sep, 23
Tool comparison	Sep, 26	Sep, 28
Implementation	Oct, 12	Dec, 6
Modelling Functional Behaviour	Oct, 12	Oct, 18
Generating Functional Tests	Oct, 19	Oct, 24
Implementing Algorithm	Oct, 25	Nov, 22
Testing with Tool A	Nov, 23	Nov, 29
Testing with Tool B	Nov, 30	Dec, 6
Evaluation	Dec, 7	Jan, 23
Defining Evaluation Strategy	Dec, 7	Dec, 13
Generating Test Cases	Dec, 14	Dec, 20
Preparing Simulation Environment	Dec, 21	Jan, 9
Running Simulations	Jan, 10	Jan, 16
Evaluating Results	Jan, 17	Jan, 23
Written Thesis	Sep, 29	Mar, 3
Researching Related Work	Sep, 29	Oct, 12
Writing State of the Art Chapter	Oct, 13	Oct, 17
Writing Introduction	Oct, 18	Oct, 24
Describing Methodology	Dec, 14	Dec, 20
Charting Simulation Results	Jan, 24	Jan, 30
Writing Evaluation Chapter	Jan, 31	Feb, 6
Writing Remaining Chapters	Feb, 7	Mar, 3



5.4 Contingency plan

Possible risks (e.g., no available resources, no sufficiently efficient algorithm exists) that can already be estimated before starting the thesis, as well as possible contingency plans. **(0,5 Pages)**

1. Risk ...

   ...

2. Risk ...

   ...